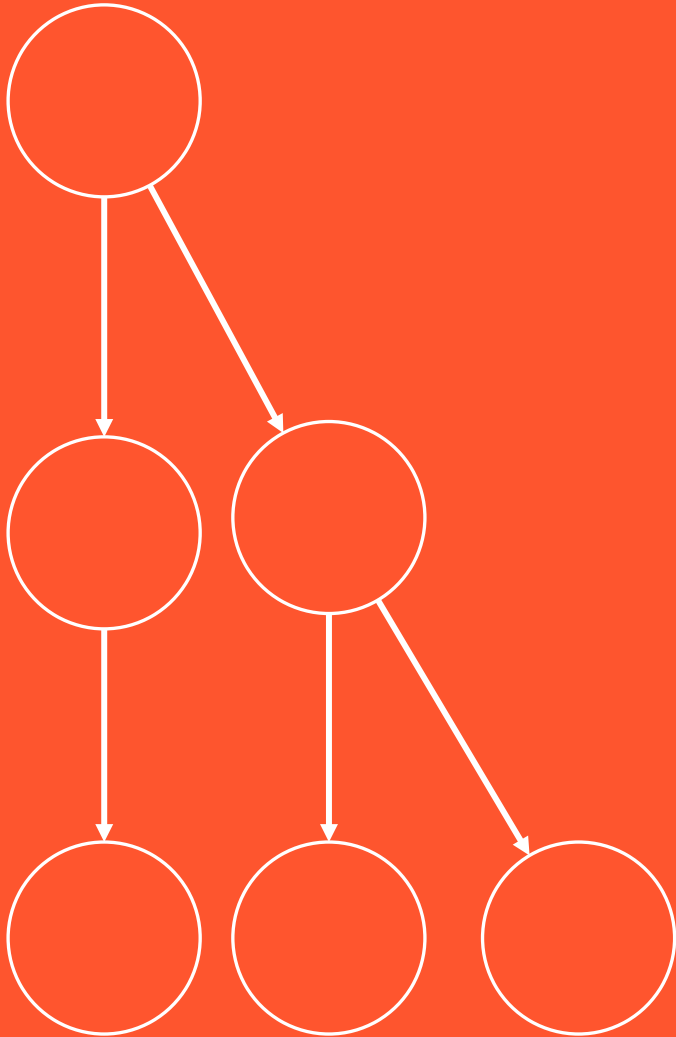




BCOG 100

Neurons and Neural Communication
Lecture 3.2 — How Neurons Process Information

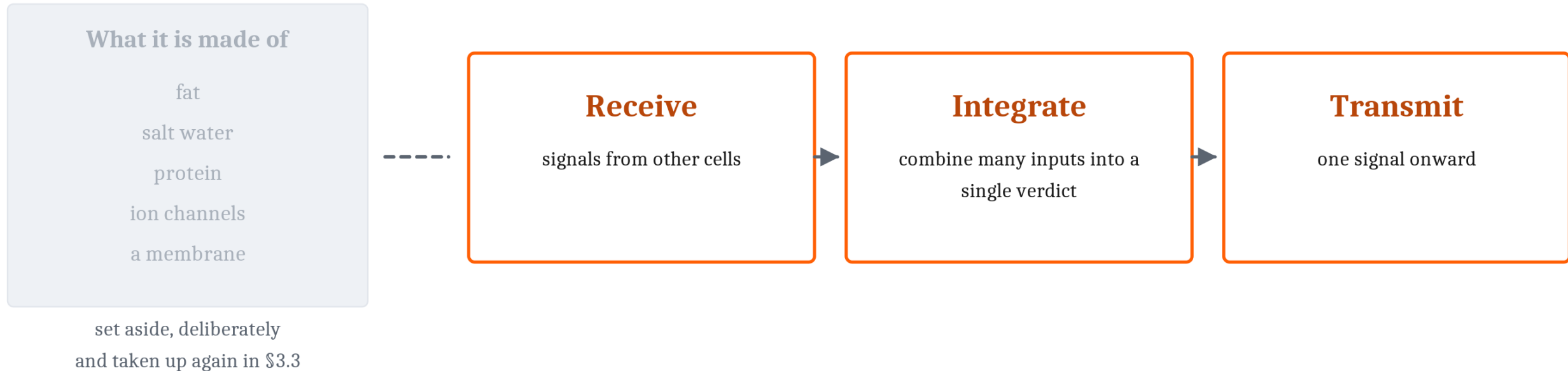


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The Algorithmic Question

For this lecture, we set the biology aside. We describe the neuron as though we did not know what it was built from. **That is deliberate, and it happens in every module from here on.**

Set the biology aside, and describe what the cell does



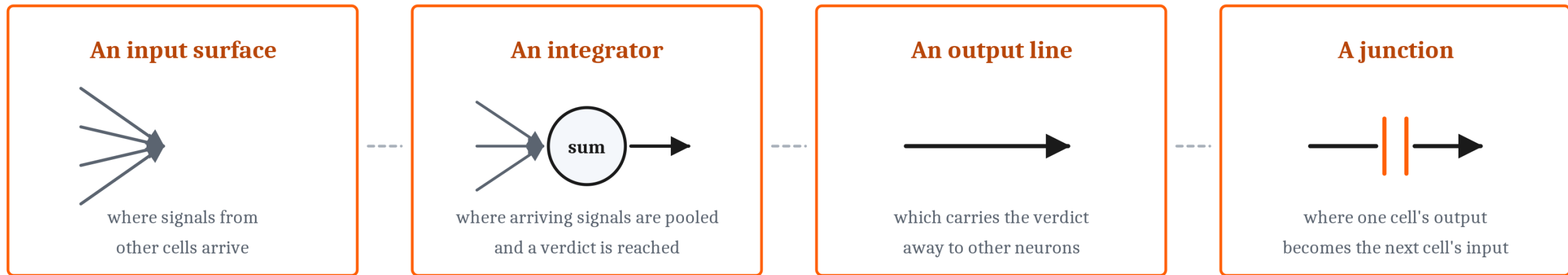
Everything else at this level is detail fleshing out those three verbs.

The same move happens in every module from here on: a functional description first, a physical one after.

Four Jobs, and Not One of Them Is a Part

Receiving, integrating and transmitting require **four things**, and at this level they are better thought of as jobs than as objects. **Each is a real piece of a real cell with a name and a shape — and the names belong to the next lecture.**

Four jobs the cell has to fill



Jobs, not parts.

Each of the four is a real piece of a real cell, with a name and a shape.

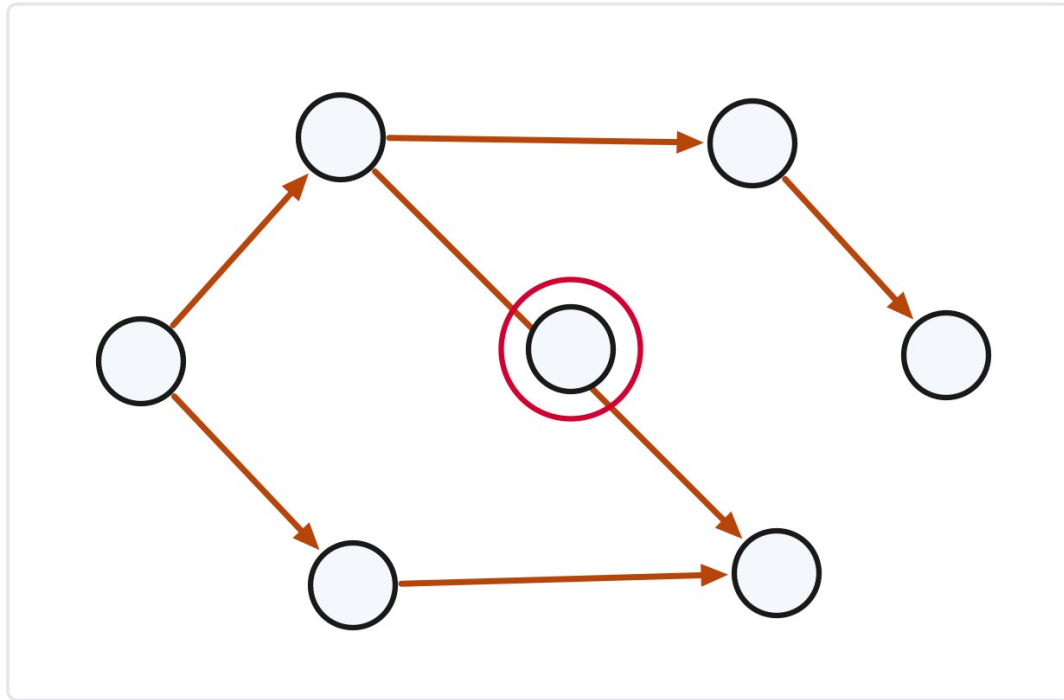
The names belong to the implementational level, and we will take them up there.

The flow within one cell is one-way, and that is what matters more than any of the four.

A Circuit, Not a Puddle

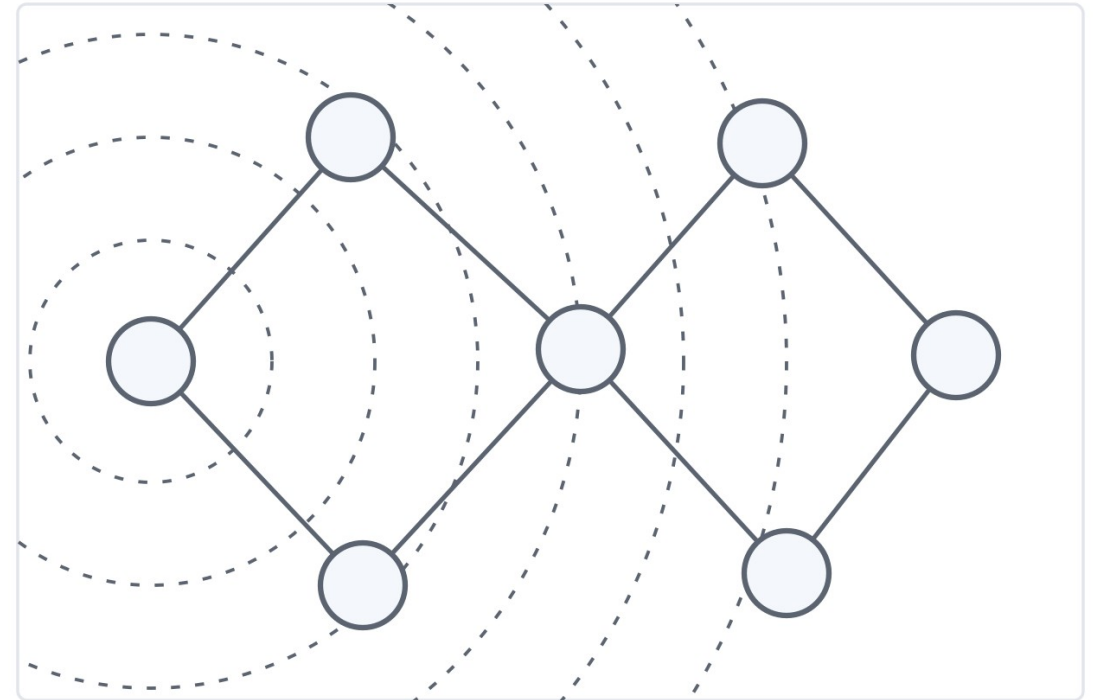
The **flow within a single neuron is one-way**, which is what makes it possible to build a circuit rather than a puddle. So connection is a matter of **wiring rather than proximity** — and in a puddle, being next to something is the same as being connected to it.

wired — one direction, particular cells



the circled cell is the nearest cell to three of these, and hears from none of them

a puddle — every neighbor, every direction



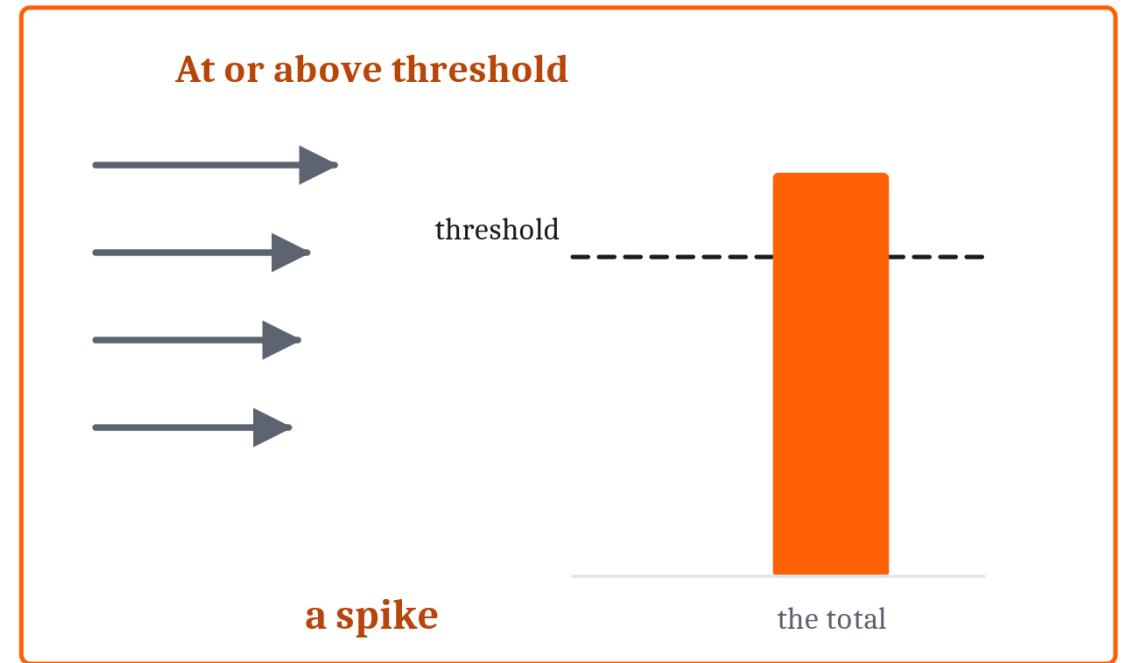
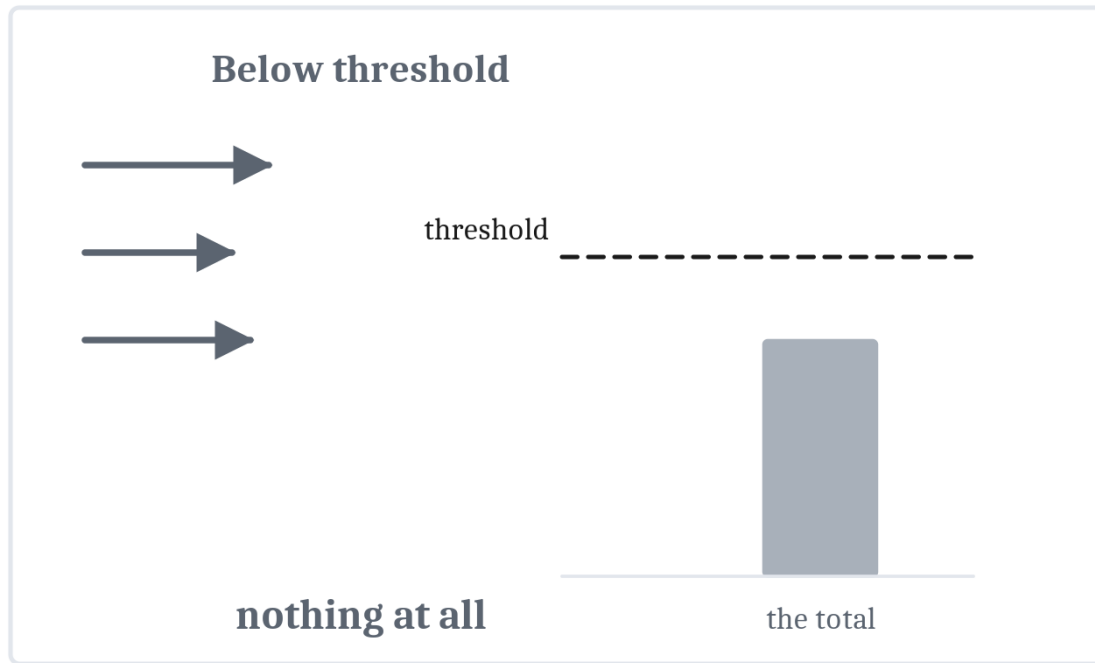
activity spreads outward from wherever it began

Same cells, same positions in both panels. Only what joins them differs.

Integration and Threshold

Integration is the middle verb: the arriving signals are added up. And then the crux — the neuron does not report the total, it compares the total against a **threshold**. Nothing in that operation requires any idea what the inputs mean.

Sum the inputs. Compare against a level. Act, or do nothing.



Not a wire and not a relay. A small evaluator, pooling evidence from many sources

and answering one question: given everything arriving right now, do I act now, or not?

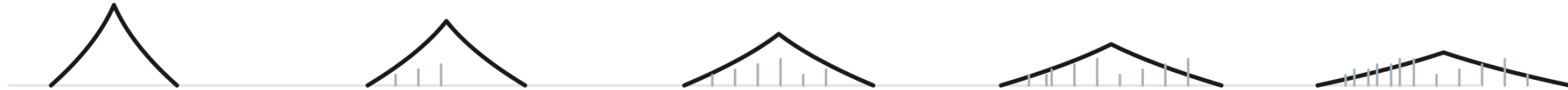
The All-or-Nothing Spike

When a neuron acts it produces an **action potential**, commonly called a *spike*. What matters at this level is its shape — which is to say that it does not *have* a variable one. **All-or-nothing**: no small ones, no large ones, none that fade partway along.

Why throw away the obvious option?

A signal whose SIZE means something

fades, blurs, picks up noise
over a meter of animal



how much of this was the message?

An all-or-nothing signal

regenerated at full strength
the whole way



what arrives is exactly what left

distance traveled

A stronger stimulus does not produce a bigger spike. Every spike a given neuron fires is essentially identical to every other one.

The Same Solution, Found Three Times

Human engineers made the same choice, independently and for the same reason. And life had made it long before neurons or electronics.

Three systems, the same problem, the same solution

A neuron



a spike, or no spike

no small ones, no large ones, none that fade
partway

Digital electronics



fully on, or fully off

a value that is only ever on or off can be
cleaned up and passed along

DNA



which of four bases sits at each position

not how much of anything is present, which
is why it survives copying

A discrete signal can be checked and restored. A continuous one can only be approximated.

Life found this solution long before neurons, and engineers found it again without asking biology.

If Every Spike Is Identical, Where Is the Message?

A neuron's message is not in any one spike, any more than the meaning of a sentence is in any one word. It is in how many, and when — the neuron's **firing rate**.

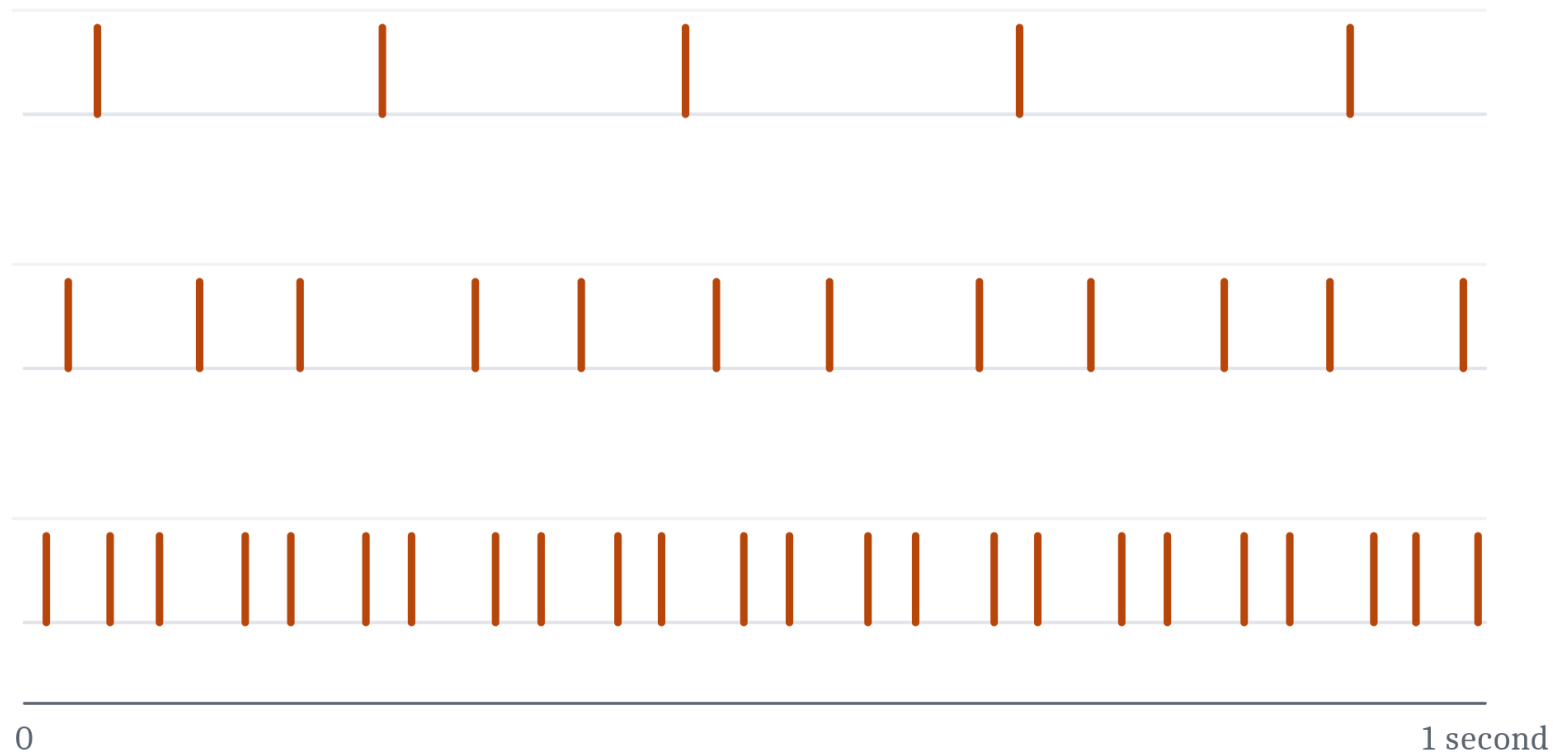
A stronger stimulus does not produce a bigger spike. It produces more spikes, faster.

pressure on the fingertip

a light touch
5 spikes

a firmer press
12 spikes

pressing hard
24 spikes



every mark is the same size — only how many, and how close together, changes

Two Ways to Describe One Output

A neuron's output is all-or-nothing. A neuron's output is a rate that can take many values. **Both are true**, and they are descriptions at two different time scales rather than competing claims.

zoomed out — a hundred milliseconds

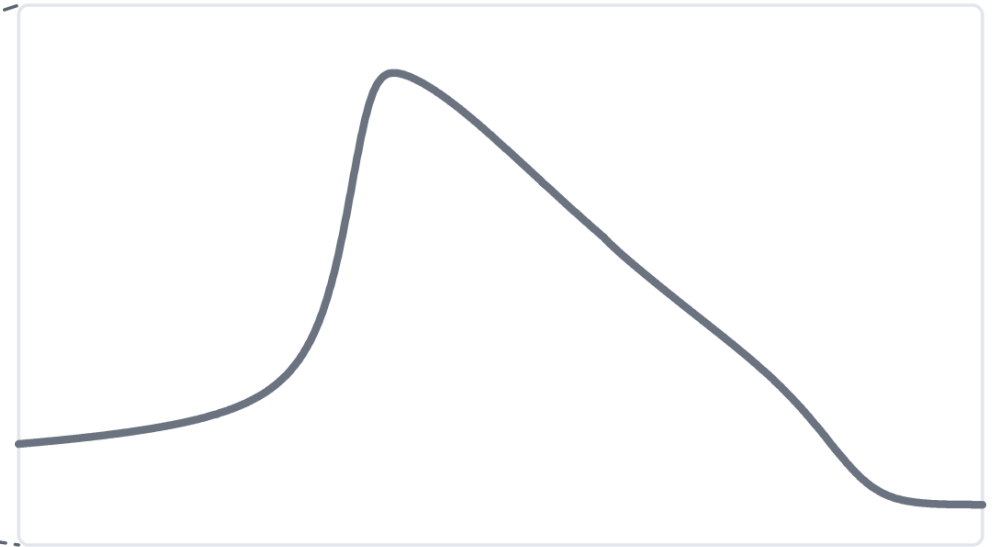
a count, and a count can be any number in a range



9 spikes in 100 ms = 90 per second

zoomed in — one millisecond

an event, and it is there or it is not



the same size every time

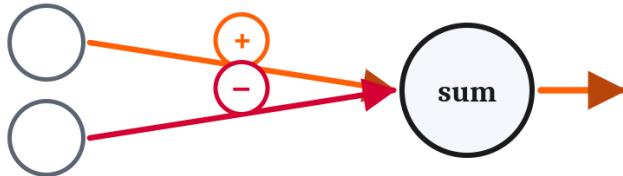
one spike train, two magnifications — not two competing claims

Excitation, Inhibition, and Modulation

So far every input has pushed the same way. That was a simplification. An arriving spike releases a **neurotransmitter: excitatory** ones push the receiving cell toward its threshold, **inhibitory** ones push it away. *Which chemicals do which job belongs to the next lecture.*

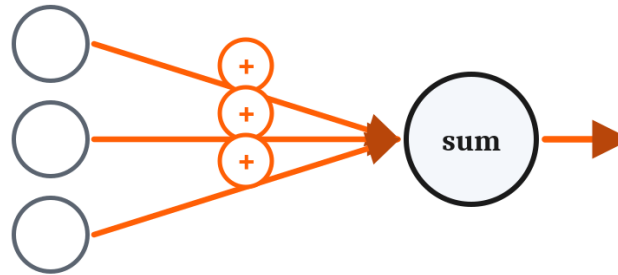
Sum two kinds of input, and the cell stops being a relay

this and not that



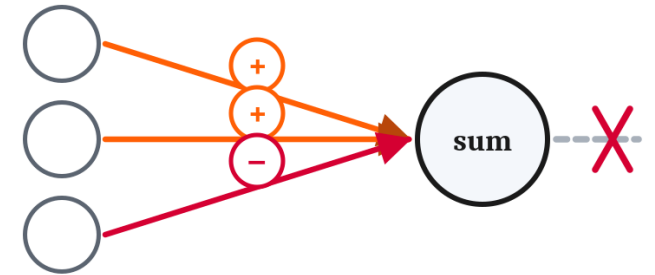
fires only when one input is active
and another is silent

a coincidence detector



stays quiet unless several inputs
arrive close together in time

a veto



shut down entirely by a single
well-placed inhibitory connection

None of these is relaying anything.

The balance between excitation and inhibition is what turns a chain of cells passing messages along into an assembly that computes something.

Writing a Neuron Down

Words are where an account like this stops in many introductory courses. **A description in words can be agreed with. A description precise enough to compute with makes predictions** — including about cases nobody has tested. That is Module 2's synthetic method: understanding by building.

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_3$$

each input's rate, times how strongly and in which direction it connects, plus the baseline

baseline $b_0 = 5$ spikes per second · an excitatory input, weight $+2$ · an inhibitory input, weight -3

$$5 + 2(10) + (-3)(0) = 25$$

the excitatory input at 10, the inhibitory one silent

$$5 + 20 - 15 = 10$$

the inhibitory input comes on at 5 — and the excitatory drive has not changed at all

$$5 + 20 - 45 = -20$$

the inhibitory input at 15 — and a neuron cannot fire a negative number of times per second

Inhibition doing arithmetic — and then the model showing us its own limitation.

So the sum cannot be the final answer. It has to pass through one more step, and that step is Module 4's.

What the Model Is Worth

A model is a deliberate simplification, and the honest way to present one is to say what it leaves out. **The way a model simplifies is a theoretical claim about which properties matter, and which do not.**

A model is wrong on purpose, in stated respects

What it leaves out

timing, entirely — a hundred spikes are a hundred spikes
the assumption that inputs simply add and subtract
no refractory period
no history
no noise

What that buys

an account precise enough to predict what the cell will do
for input combinations no one has tried

and simple enough that thousands of these units can be
wired together and the whole assembly worked out

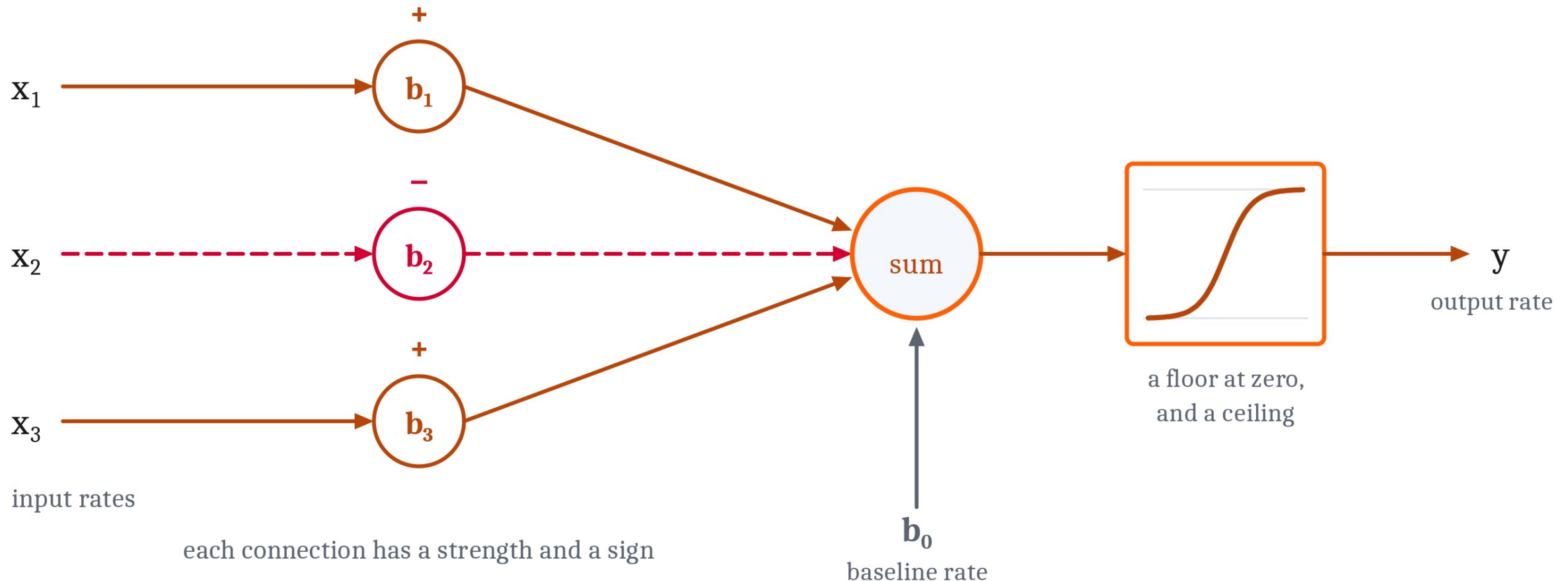
The way a model simplifies is a theoretical claim about which properties matter, and which do not.

Whether these particular claims are right is an open, empirical question.

Some researchers argue the omissions are too important: that a single cortical neuron is better understood as a small network than as a single unit, and that reproducing one can take an artificial network several layers deep.

The Same Function in Silicon

Numbers in. Each multiplied by a weight. Summed. Passed through a function. **That is the model of a biological neuron we arrived at by describing a cell — and it is also, almost exactly, the unit at the heart of an artificial neural network.**



From One Neuron to a Behaving Animal

Caenorhabditis elegans: a soil worm about a millimeter long, with exactly **302 neurons**, and the first animal whose entire wiring diagram was worked out — a map we call a **connectome**.

Sense, integrate, act — in an animal, using three or four cells in a row.

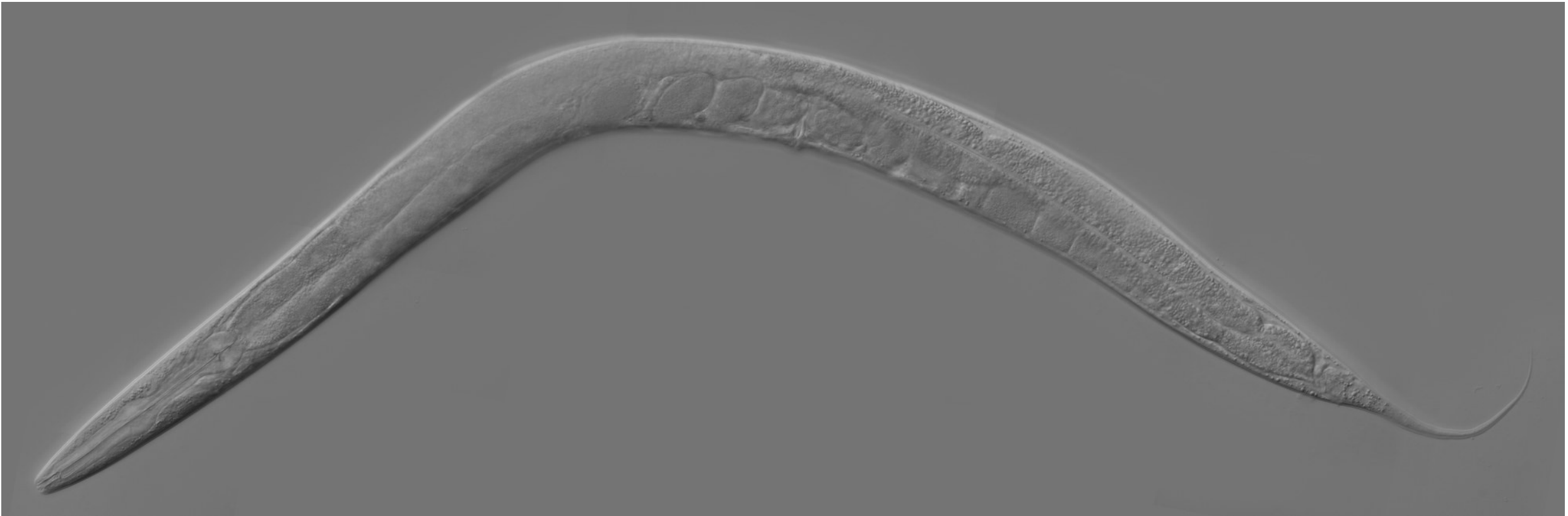


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12	What it leaves out, and what that buys	fig-model-omissions.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
13	A single unit, biological or artificial	fig-neuron-model.png	CC BY 4.0	Jon Willits — the chapter's own figure, from figures/03_neurons_and_neural_communication/make_figures.py · own work
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